PGCP-AC AUGUST 26
Assignment No- 4

Write Create a Employee ,Department & Project Tables .Add atleast 5 records in each table & Write a following Queries.

1. Write a Relational Algebra expression to perform **Cartesian Product (Cross Join)** between Emp and Dept tables.
   Answer:
```
mysql> SELECT * FROM EMP CROSS JOIN DEPT;
```

2. Write a SQL query to perform an **INNER JOIN** between Emp and Dept tables displaying EmpName and DeptName.
   Answer:
```
mysql> SELECT E.NAME,D.DNAME FROM EMP E INNER JOIN DEPT D ON E.DEPT_ID = D.DEPT_ID;
```

3. Write a SQL query to perform a **LEFT OUTER JOIN** between Emp and Dept tables to display all employees along with their department information, including employees without a department.
   Answer:

```
mysql> SELECT E.* , D.* FROM EMP E LEFT OUTER JOIN DEPT D ON E.DEPT_ID = D.DEPT_ID;
```

4. Write a SQL query to perform a **RIGHT OUTER JOIN** between Emp and Dept tables.
   Answer:
```
mysql> SELECT E.* , D.* FROM EMP E RIGHT JOIN DEPT D ON E.DEPT_ID = D.DEPT_ID;
```

5. Write a SQL query to perform a **NATURAL JOIN** between Emp and Dept.
   Answer:
```
mysql> SELECT * FROM EMP NATURAL JOIN DEPT;
```

6. Write a SQL query to perform a **CROSS JOIN** between Emp and Project table.
   Answer:

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```
mysql> SELECT * FROM EMP E CROSS JOIN PROJECT P;
```

7. Write a SQL query to **create a new table EmpBackup with the same structure as Emp but no data**.
   Answer:

```
mysql> CREATE TABLE EMPBACKUP LIKE EMP;
Query OK, 0 rows affected (0.08 sec)
```

8. Write a SQL query to **copy all data from Emp into EmpBackup table**.
   Answer:

```
mysql> INSERT INTO EMPBACKUP SELECT * FROM EMP;
Query OK, 10 rows affected (0.02 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

9. Write a SQL query to **create a new table ProjectArchive with the same structure and data as Project**.
   Answer:

```
mysql> CREATE TABLE PROJECTARCHIVE SELECT * FROM PROJECT;
```

10. Write a SQL query to create an **AUTO_INCREMENT sequence for EmpID in the Emp table** during table creation.
    Answer:

```
mysql> ALTER TABLE EMP MODIFY EMP_ID INT AUTO_INCREMENT;
Query OK, 10 rows affected (0.15 sec)
Records: 10  Duplicates: 0  Warnings: 0
```

11. Write a SQL query using a **subquery** to find all employees whose salary is greater than the **average salary of all employees**.
    Answer:

```
mysql> SELECT * FROM EMP WHERE SALARY > (SELECT AVG(SALARY) FROM EMP);
```

12. Write a SQL query using a **subquery in WHERE clause** to find employees working in departments located in 'Mumbai'.
    Answer:

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```
mysql> SELECT * FROM EMP WHERE DEPT_ID = ANY(SELECT DEPT_ID FROM DEPT WHERE LOCATION = "MUMB
AI");
+--------+--------+----------+---------+---------+------+
| EMP_ID | NAME   | SALARY   | JOINING | DEPT_ID | NOTE |
+--------+--------+----------+---------+---------+------+
|    106 | Neha   | 60500.00 |    2021 |       1 | NULL |
|    110 | Anjali | 63800.00 |    2020 |       1 | NULL |
+--------+--------+----------+---------+---------+------+
2 rows in set (0.01 sec)
```

13. Write a SQL query to display all departments where the number of employees is greater than **2**, using a subquery.

    Answer:

```
mysql> SELECT D.DNAME,COUNT(E.EMP_ID) FROM DEPT D JOIN EMP E ON D.DEPT_ID = E.DEPT_ID WHERE
D.DEPT_ID IN (SELECT DEPT_ID FROM EMP GROUP BY DEPT_ID HAVING COUNT(EMP_ID)>2)GROUP BY D.DEP
T_ID;
+-------+-----------------+
| DNAME | COUNT(E.EMP_ID) |
+-------+-----------------+
| HR    |               3 |
+-------+-----------------+
1 row in set (0.00 sec)
```

14. Write a subqueries for find out Max salary of employee.

    Answer:

```
mysql> SELECT SALARY FROM EMP WHERE SALARY = (SELECT MAX(SALARY) FROM EMP);
+----------+
| SALARY   |
+----------+
| 65000.00 |
+----------+
1 row in set (0.00 sec)
```

15. Write a self join query for each table.

    Answer:

```
mysql> SELECT * FROM EMP E JOIN EMP D ON E.DEPT_ID = D.DEPT_ID;
```

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